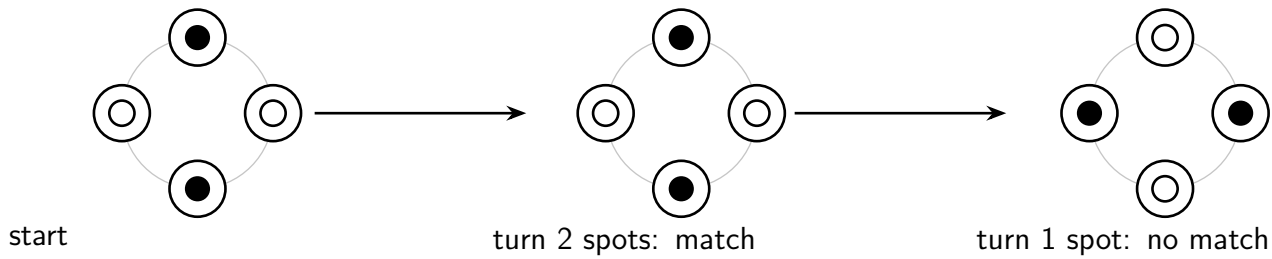
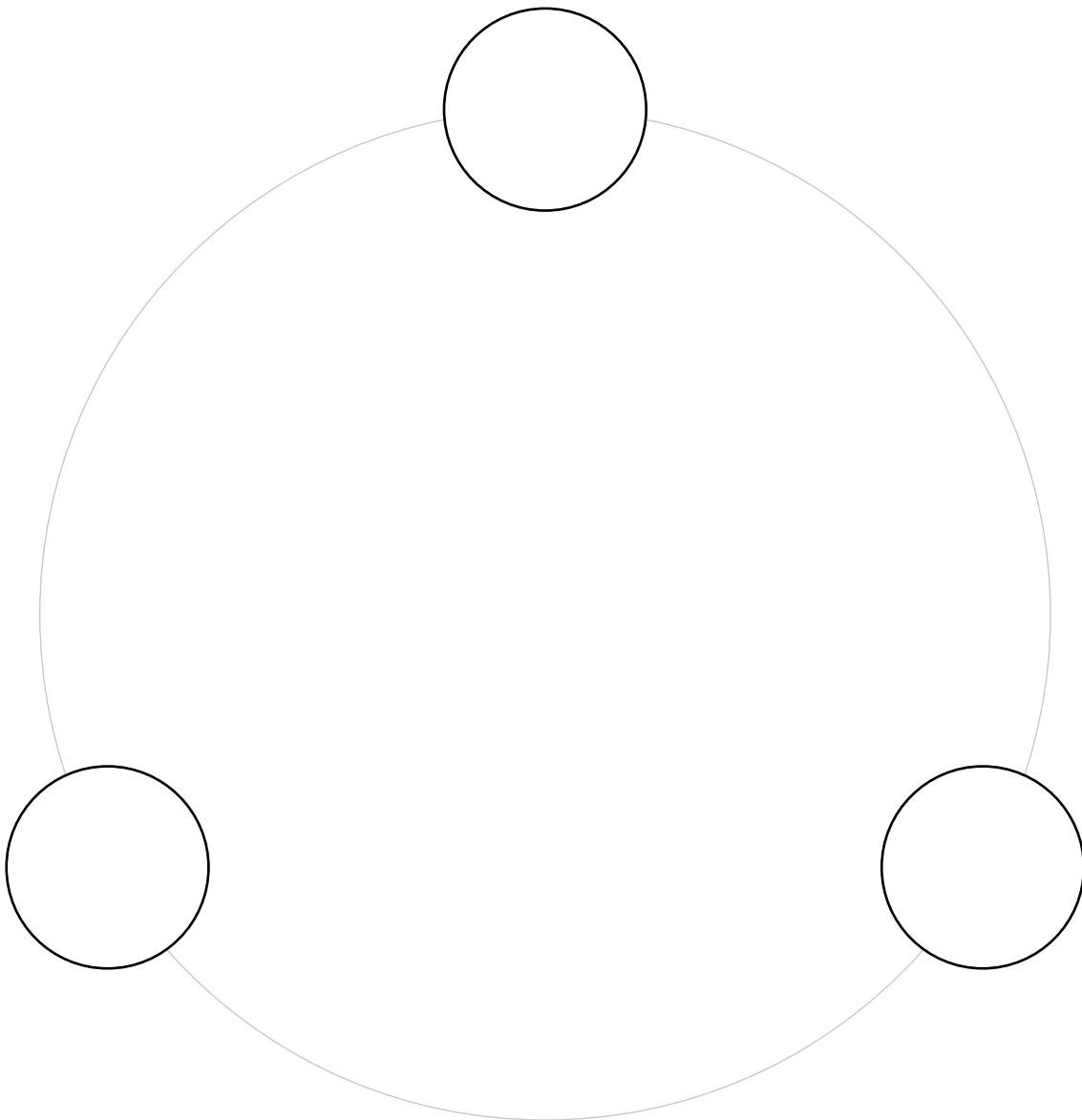


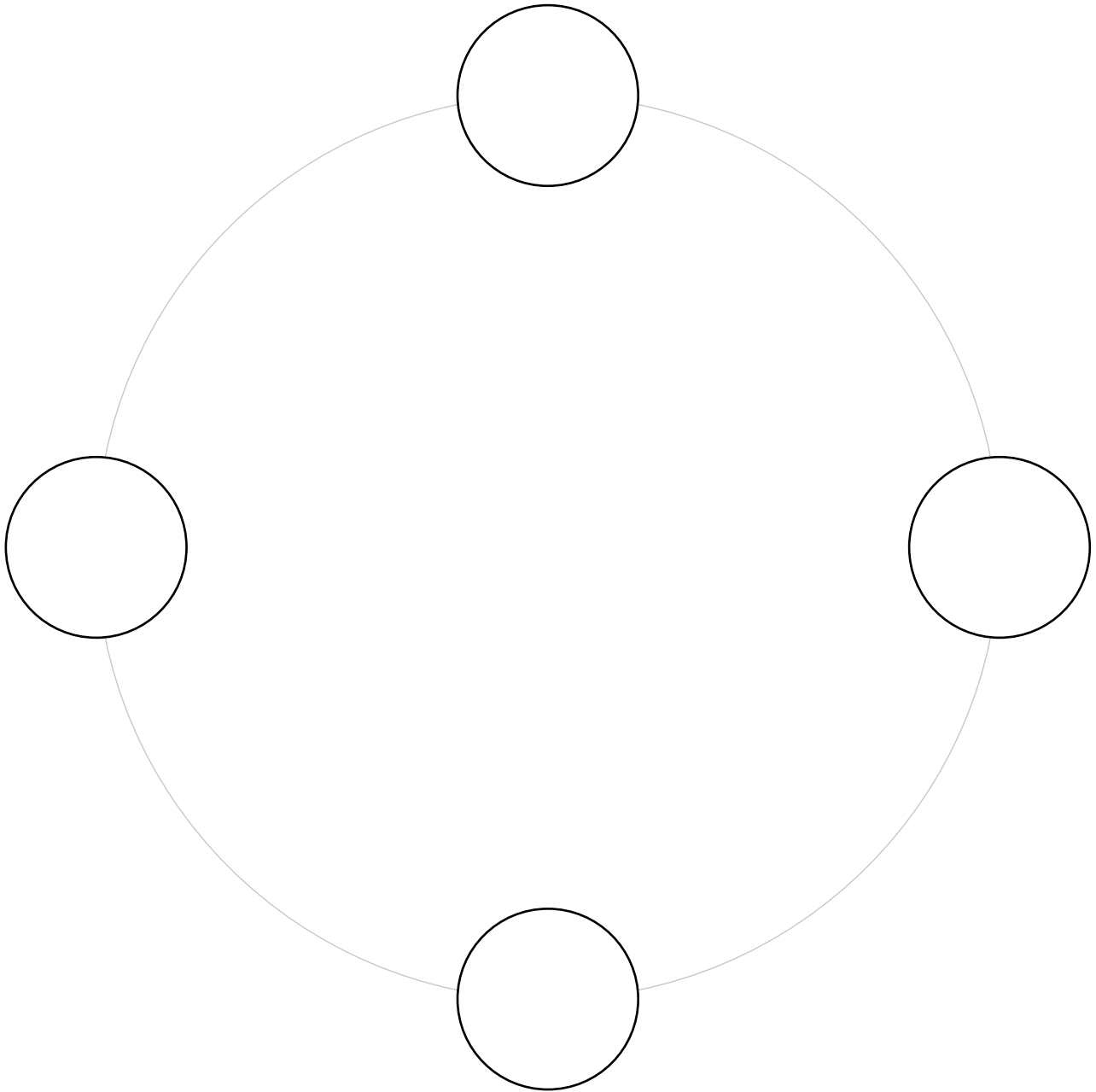
Fill every spot. Only counter kinds and spots count; the direction of a symbol does not. Keep the original still. Turn or turn over its tracing copy until all spots line up. A whole turn does not count.



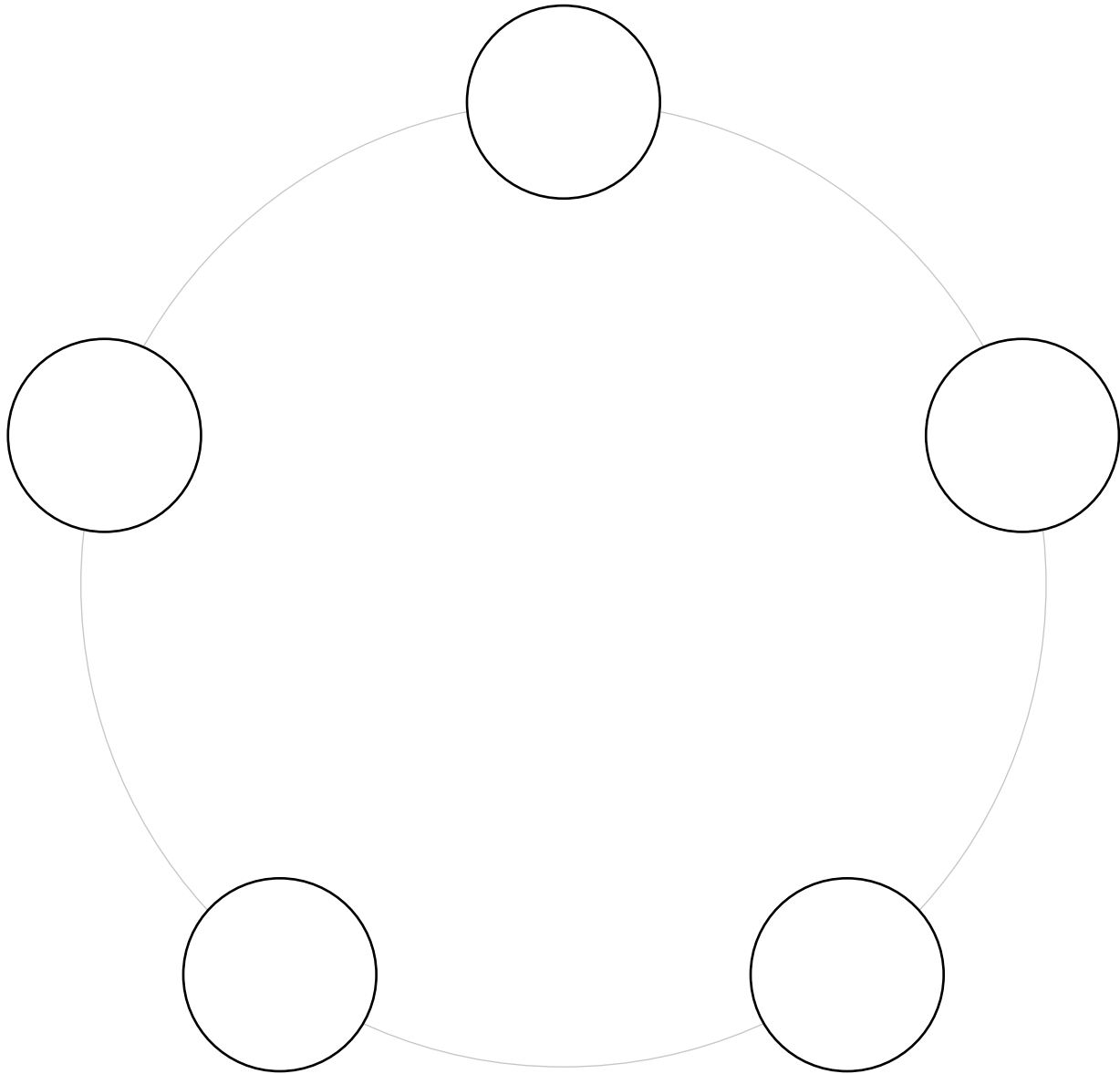
Problem 1: Find the fewest kinds of counter that stop every turn and flip on the three-ring. Neighbors may match. Can you rule out every pattern with fewer kinds?



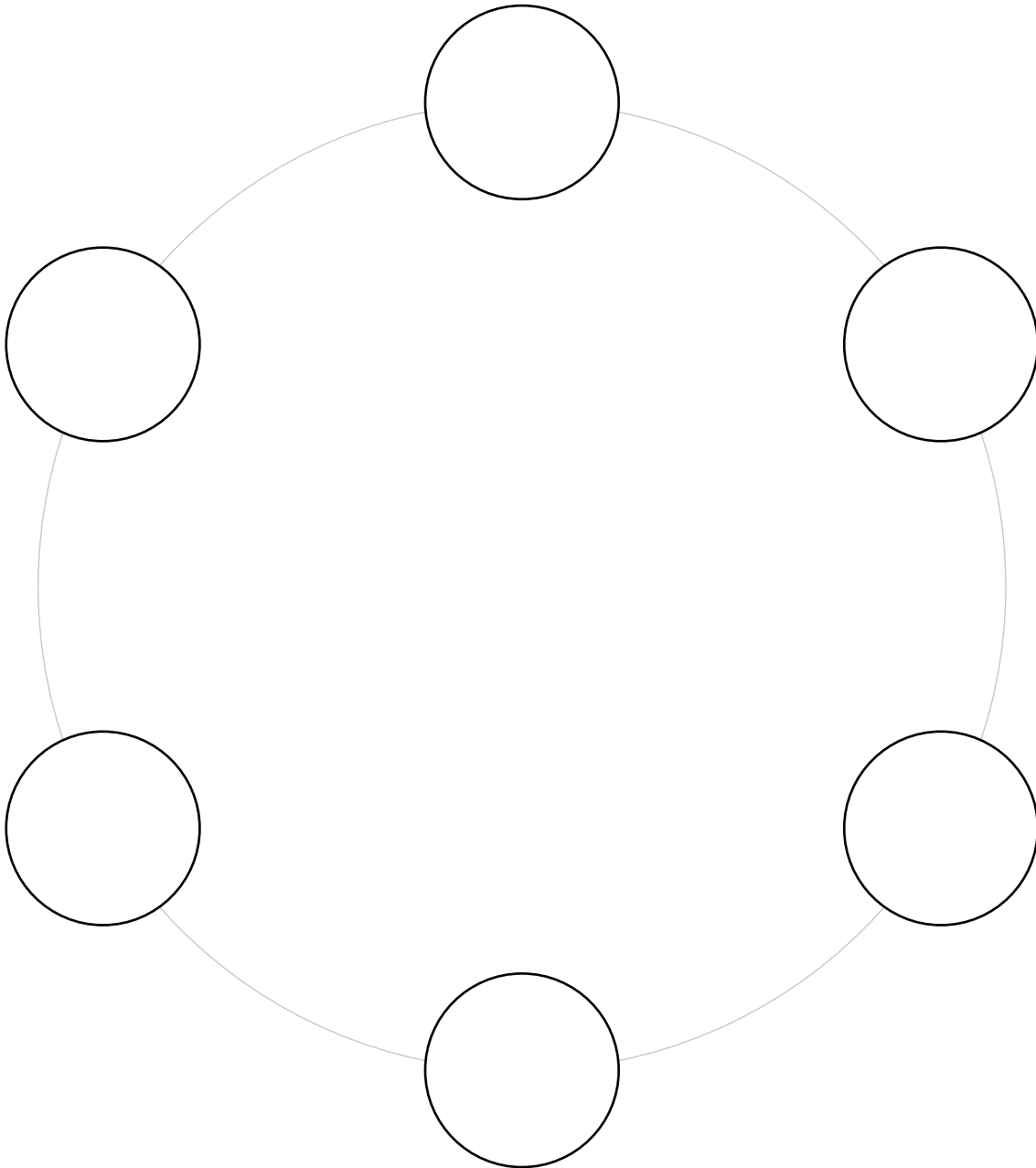
Problem 2: Find the fewest kinds of counter that stop every turn and flip on the four-ring. Explain why fewer kinds cannot work.



Problem 3: Find the fewest kinds of counter that stop every turn and flip on the five-ring. Find an explanation that also works for the three-ring and four-ring.



Problem 4: Use two kinds of counter to stop every turn and flip on this six-ring. Explain why no turn or flip can match it.



Problem 5: Can you make a two-kind pattern that stops every turn and flip on any ring with at least six spots? Test your idea on this eight-ring, then explain why it always works.

